Exact paving for Fubini products of normal measures, with an application to maximal projection filters*

David V. Feldman† Alexander Wilce‡

Abstract

Let κ be measurable and $\mathcal{D}$ a normal measure on κ. We prove that the Fubini product $\mathcal{D} \otimes \mathcal{D}$ satisfies *exact paving*: for every assignment of a countable set c_x to each point x, some set of $\mathcal{D} \otimes \mathcal{D}$ mutually avoids the assignment. The proof uses only Rowbottom homogeneity for finitely many colorings of triples and quadruples; normality of the product is neither available nor needed. Consequently the block filter of $\mathcal{D} \otimes \mathcal{D}$ in the projection lattice of $\ell^2(\kappa \times \kappa)$ is a κ-complete nonprincipal maximal filter with a singleton state face — although $\mathcal{D} \otimes \mathcal{D}$ is not isomorphic to any normal measure. This answers, negatively, the question of the third paper in this series: maximality of the block filter does not characterize normality up to isomorphism. We organize the invariants that do and do not survive: the full-selector property of normal measures, which holds for arbitrary partitions, is isomorphism-invariant, and fails for the product; the σ-Q-point property, which the product enjoys, by an injectivity-on-a-square theorem; and an exact-paving property lying between them. In a final section we return to $\aleph_0$ and the rigidity conjecture of the second paper, proving three structural constraints on any addable blocker of the witness subspace: it meets the block of any m pieces in dimension at most m, its total support belongs to the ultrafilter, and on every measure-one block it contains vectors of infinite piece-spread.

1 Introduction

This paper continues a series [1, 2, 3] on maximal filters in the projection lattice $\mathcal{P}(H)$ of a real Hilbert space (*plufs*), the lattice-theoretic linearization of "bounded sequences converge along ultrafilters." The third paper established that at a measurable cardinal κ, along a normal measure $\mathcal{D}$, the theory collapses to an exact form: every bounded operator is exactly diagonal on a measure-one coordinate block (*exact paving*), every closed subspace either contains a measure-one block or meets one trivially (*exact dichotomy*), and consequently the block filter

$$\Phi(\mathcal{U}) \;=\; \{M \in \mathcal{P}(\ell^2(\kappa)) : \mathcal{H}_S \subseteq M \text{ for some } S \in \mathcal{U}\}$$

is a κ-complete nonprincipal pluf with a singleton state face. It also proved a necessity result: maximality of $\Phi(\mathcal{U})$ forces $\mathcal{U}$ to be a σ-*Q-point* (every partition of κ into countable pieces admits a partial selector in $\mathcal{U}$), a property enjoyed by every normal measure, and asked ([3], Question

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†University of New Hampshire; dvfinnh@gmail.com.

‡Susquehanna University.

5.5) whether maximality characterizes the normal measures up to isomorphism, proposing the Fubini product $\mathcal{D} \otimes \mathcal{D}$ as the test case.

The present paper settles the test case, and with it the question. The product is not isomorphic to any normal measure (Proposition 2.3), yet its block filter is maximal (Theorem 5.2). The engine is Theorem 4.1: $\mathcal{D} \otimes \mathcal{D}$ satisfies exact paving outright. Normality entered the paving proof of [3] through the diagonal intersection; the product admits no diagonal intersections, but it inherits from $\mathcal{D}$ something just as serviceable, Rowbottom homogeneity for finite colorings of finite subsets of κ, and homogeneity suffices: any assignment of countable avoidance-sets, examined through the finitely many *membership patterns* it can exhibit on triples and quadruples, is forced by countability to avoid a homogeneous square.

The same method yields an injectivity theorem (Theorem 3.1) — every countable-to-one function on pairs from κ is injective on the pairs from a $\mathcal{D}$-large set — which gives the σ-Q-point property of the product directly.

Since the mechanized development accompanying [3] consumed normality only through the paving conclusion, the entire positive theory transfers to the product along Theorem 4.1; we spell this out in Section 5, introduce the *exact-paving property* (EPP) as the invariant the transfer isolates, and record the resulting picture:

$$\text{normal} \implies \text{EPP} \implies \Phi \text{ maximal} \implies \sigma\text{-Q-point},$$

so that maximality holds strictly beyond the normal measures. What *does* separate normal measures from products is a selector property: normal measures admit full selectors for *arbitrary* partitions with no piece in the measure (Lemma 2.2), the piece-minima argument of [3] requiring no countability. The property is isomorphism-invariant, and the column partition refutes it for the product. The invariants thus separate: $\mathcal{D} \otimes \mathcal{D}$ is maximal and σ-Q but not Q-for-all-partitions.

Section 6 returns to $\aleph_0$ and the rigidity conjecture of [2]: for a non-selective ultrafilter $\mathcal{U}$ on $\mathbb{N}$ with witness subspace W, does W belong to every pluf extension of the block filter? We prove three constraints on any prospective *addable blocker* R (a closed subspace with $R \cap W = 0$ meeting every member of the block filter in infinite dimension): a generalized thinness bound (Lemma 6.1), a support lemma (Lemma 6.2), and a Baire-category piece-spread lemma (Lemma 6.3). The conjecture remains open; the constraints close every finitely-generated route to a counterexample. The measurable-cardinal results of this paper frame the conjecture from above: along $\mathcal{D}$ or $\mathcal{D} \otimes \mathcal{D}$ there are no witness subspaces at all, so rigidity, if real, is a phenomenon of small cardinality.

Throughout, κ is a measurable cardinal, $\mathcal{D}$ a normal measure (nonprincipal κ-complete normal ultrafilter) on κ, and $\mathcal{D} \otimes \mathcal{D}$ the Fubini product on $\kappa \times \kappa$: $X \in \mathcal{D} \otimes \mathcal{D}$ iff $\{\gamma : \{\beta : (\gamma, \beta) \in X\} \in \mathcal{D}\} \in \mathcal{D}$. We write $[A]^2$ for the two-element subsets of A, identified when convenient with the ordered pairs (γ, β), $\gamma < \beta$; since each section $\{\beta : \beta > \gamma\}$ lies in $\mathcal{D}$, the product concentrates on the upper triangle, and we pass between the triangle and $[\kappa]^2$ without comment. The Hilbert-space background and the definitions of pluf, block, block filter and witness subspace are as in [1, 2, 3].

2 Homogeneity preliminaries

We use one classical input.

Theorem 2.1 (Rowbottom; see [5, Theorem 7.17])**.** *Let $\mathcal{D}$ be a normal measure on κ, $n < \omega$, $\lambda < \kappa$, and $F : [\kappa]^n \to \lambda$. Then there is $H \in \mathcal{D}$ with F constant on $[H]^n$.*

Applying Theorem 2.1 to the finite product of finitely many colorings homogenizes them simultaneously; we use this silently. The following one-line observation supplies all the "high pivots" below: if $H \in \mathcal{D}$ then H has order type κ, so for every $\lambda < \kappa$ some (indeed, a tail of) $\beta \in H$ satisfies $|H \cap \beta| \geq \lambda$.

The selector lemma of [3] holds for arbitrary partitions: countability of the pieces plays no role in its proof.

Lemma 2.2 (Full selectors along normal measures)**.** *Let $\mathcal{D}$ be normal and $\kappa = \bigsqcup_{i \in I} \sigma_i$ a partition into nonempty pieces with $\sigma_i \notin \mathcal{D}$ for every i. Then the set $M = \{\min \sigma_i : i \in I\}$ of piece-minima lies in $\mathcal{D}$; in particular $\mathcal{D}$ contains a set meeting every piece exactly once.*

Proof. If $M \notin \mathcal{D}$ then on $\kappa \setminus M \in \mathcal{D}$ the map $g(\alpha) = \min(\text{piece of } \alpha)$ is regressive, hence constant on a set of $\mathcal{D}$ by normality; that set lies inside a single piece, contradicting $\sigma_i \notin \mathcal{D}$. So $M \in \mathcal{D}$, and M meets each piece exactly once by disjointness of the pieces and distinctness of their minima. □

The property in the conclusion — every partition with no piece in the ultrafilter admits a selector in the ultrafilter — mentions no order and is therefore invariant under isomorphism of ultrafilters (transport along a bijection of the underlying sets). We refer to it as the *full-selector property.* The countable-piece specialization is the σ-Q-point property of [3].

Proposition 2.3. *$\mathcal{D} \otimes \mathcal{D}$ fails the full-selector property; consequently $\mathcal{D} \otimes \mathcal{D}$ is not isomorphic to any normal measure.*

Proof. Partition $\kappa \times \kappa$ into the columns $C_\gamma = \{\gamma\} \times \kappa$. No column lies in $\mathcal{D} \otimes \mathcal{D}$ (the set of first coordinates with $\mathcal{D}$-large section is a singleton, not in $\mathcal{D}$). A selector meets each column at most once, so its sections are singletons, none in $\mathcal{D}$; hence no selector lies in $\mathcal{D} \otimes \mathcal{D}$. Since normal measures have the full-selector property (Lemma 2.2) and the property is isomorphism-invariant, no isomorph of a normal measure can be $\mathcal{D} \otimes \mathcal{D}$. □

3 Injectivity on a square

Theorem 3.1. *Let $\mathcal{D}$ be a normal measure on κ, $\lambda < \kappa$, and let $g : [\kappa]^2 \to V$ be a function all of whose fibers have size less than $\max(\lambda, \omega_1)$. Then there is $H \in \mathcal{D}$ with g injective on $[H]^2$.*

Proof. Color the three-element subsets $\{h_1 < h_2 < h_3\}$ of κ by the equality pattern of g on their three pairs, and the four-element subsets $\{h_1 < h_2 < h_3 < h_4\}$ by the equality pattern of g on their six pairs; each coloring has finitely many values. By Theorem 2.1, fix $H \in \mathcal{D}$ homogeneous for both. Write $\mu = \max(\lambda, \omega_1)$; all fibers of g have size $< \mu$, and by the order-type observation of Section 2 we may pick elements of H with at least μ predecessors in H whenever needed.

We claim the homogeneous patterns identify no two pairs; injectivity on $[H]^2$ follows, since two distinct pairs from H span a three- or four-element subset of H. Suppose some identification held homogeneously; we derive in each case a fiber of g of size $\geq \mu$.

Three-element patterns. If $g(h_1, h_2) = g(h_1, h_3)$ homogeneously, fix $h_1 < h_2$ in H and vary h_3 over the tail of H above h_2: the fiber of $g(h_1, h_2)$ contains κ pairs. If $g(h_1, h_2) = g(h_2, h_3)$, fix $h_1 < h_2$ and vary h_3: again κ pairs. If $g(h_1, h_3) = g(h_2, h_3)$, choose $h_2 < h_3$ in H with $|H \cap h_2| \geq \mu$ and vary h_1 over $H \cap h_2$: the fiber of $g(h_2, h_3)$ contains μ pairs.

Four-element patterns (identifications between the two coordinate-disjoint pairs; identifications sharing a coordinate reduce to three-element patterns). If $g(h_1, h_2) = g(h_3, h_4)$, fix $h_1 < h_2$ and vary $h_3 < h_4$ above h_2: κ pairs in one fiber. If $g(h_1, h_3) = g(h_2, h_4)$, fix $h_1 < h_2 < h_3$ and vary h_4: κ pairs share the value $g(h_1, h_3)$. If $g(h_1, h_4) = g(h_2, h_3)$, fix $h_2 < h_3$ and $h_1 < h_2$, and vary h_4: κ pairs share the value $g(h_2, h_3)$.

Each case contradicts the fiber bound; so all patterns are "all-distinct," and g is injective on $[H]^2$. $\square$

Corollary 3.2. *$\mathcal{D} \otimes \mathcal{D}$ is a σ-Q-point; indeed, for every $\lambda < \kappa$, every partition of $\kappa \times \kappa$ into pieces of size $< \lambda$ admits a partial selector in $\mathcal{D} \otimes \mathcal{D}$, of the form $\{(\gamma, \beta) : \gamma < \beta \in H\}$ with $H \in \mathcal{D}$.*

Proof. The product concentrates on the upper triangle, which we identify with $[\kappa]^2$. Let g assign to each point its piece; the fibers are the pieces, of size $< \max(\lambda, \omega_1) < \kappa$. Theorem 3.1 gives $H \in \mathcal{D}$ with g injective on $[H]^2$; the upper triangle $S = \{(\gamma, \beta) : \gamma, \beta \in H,\ \gamma < \beta\}$ lies in $\mathcal{D} \otimes \mathcal{D}$ (its section at $\gamma \in H$ is a tail of H), and distinct points of S are distinct pairs of $[H]^2$, hence lie in distinct pieces. $\square$

Together with Proposition 2.3, the corollary separates two invariants: the σ-Q-point property does not imply the full-selector property.

4 Exact paving for the product

Theorem 4.1. *Let $\mathcal{D}$ be a normal measure on κ and let each point $x \in \kappa \times \kappa$ carry a countable set $c_x \subseteq \kappa \times \kappa$. Then there is $S \in \mathcal{D} \otimes \mathcal{D}$ such that $y \notin c_x$ for all distinct $x, y \in S$. More generally the conclusion holds with "countable" replaced by "of size $< \lambda$" for any $\lambda < \kappa$.*

Proof. Write $\mu = \max(\lambda, \omega_1)$ in the general form; in the countable case $\mu = \omega_1$. As before, work on the upper triangle, identifying points with pairs. For each ordered pair of *placement patterns* on a four-element set $\{h_1 < h_2 < h_3 < h_4\}$ — a choice of two disjoint pairs from the four elements, one designated x and the other y — color the four-element sets of κ by the truth value of "$y \in c_x$"; there are finitely many placement patterns. Similarly, for each ordered pair of distinct pairs from a three-element set $\{h_1 < h_2 < h_3\}$ (patterns in which x and y share one coordinate), color the three-element sets by the truth value of "$y \in c_x$." By Theorem 2.1 fix $H \in \mathcal{D}$ homogeneous for all these finitely many colorings simultaneously, and let S be the upper triangle of H; as in Corollary 3.2, $S \in \mathcal{D} \otimes \mathcal{D}$.

Distinct points $x, y \in S$ span a three- or four-element subset of H in one of the enumerated patterns, so it suffices to show every pattern is homogeneously colored "$y \notin c_x$." Suppose some pattern were homogeneously "yes"; in each case we produce a single x with $|c_x| \geq \mu$, contradicting the size bound. The scheme is uniform: holding x's coordinates fixed at suitably chosen elements of H, the coordinates of y still range over a large set, and homogeneity puts every resulting y into the one set c_x. Each unordered placement must be treated in *both* role assignments, giving twelve patterns in all. In four of them the large family lives in the region *between* two fixed coordinates: there one passes first to the tail of H above the lower coordinate

— a set of $\mathcal{D}$ — and takes the pivot inside that tail, so that uncountably many elements of H lie strictly between.

Patterns in which y owns the top position h_4, with x on two of the remaining positions: fix x's coordinates (choosing, when the pattern requires an element of H strictly between them or below them, elements of H making those regions nonempty — possible since H is unbounded of order type κ), and vary h_4 over a tail of H: κ-many values of y, all in c_x.

Patterns in which x owns the top position and y lies among the lower positions: choose x's lower coordinate to be an element h of H with $|H \cap h| \geq \mu$ (order-type observation), and its top coordinate above it; the free coordinates of y then range over subsets of $H \cap h$ (together, when the pattern permits, with a region between x's coordinates, which only helps), producing at least μ-many values of y in c_x. Concretely: for $y = (h_1, h_2)$, $x = (h_3, h_4)$, take h_3 with μ predecessors in H and count the pairs from $H \cap h_3$; for $y = (h_1, h_3)$, $x = (h_2, h_4)$, take h_2 with μ predecessors and h_4 large, and count $h_1 \in H \cap h_2$ (with any fixed admissible h_3); for $y = (h_2, h_3)$, $x = (h_1, h_4)$, take h_4 so that $|H \cap (h_1, h_4)| \geq \mu$ and count the pairs between.

Three-element patterns: if y's non-shared coordinate can be varied upward (patterns $y = (h_1, h_3)$ or (h_2, h_3) with h_3 free above x, and the shared-coordinate analogues), fix x and vary it: κ-many y's. If y's non-shared coordinate lies below a coordinate of x (patterns $y = (h_1, h_2)$ or (h_1, h_3) with h_1 free below), choose the relevant coordinate of x with μ predecessors in H and count.

Every "yes" is thus impossible; all patterns are homogeneously "$y \notin c_x$," and S mutually avoids the assignment. $\square$

Remark 4.2. The proof uses nothing about $\mathcal{D} \otimes \mathcal{D}$ beyond: it contains the upper triangles of Rowbottom-homogeneous sets. It therefore applies verbatim to all finite Fubini powers $\mathcal{D}^{\otimes n}$ (color $2n$-element sets by membership patterns; every "yes" pattern forces a large c_x by freezing x and freeing a coordinate of y, upward or below a high pivot), and to any ultrafilter extending the filter of triangle-sets of $\mathcal{D}$-homogeneous sets. We will not need this generality.

5 Transfer, and the answer to Question 5.5 of [3]

Definition 5.1. A κ-complete nonprincipal ultrafilter $\mathcal{U}$ on a set X of size κ has the *exact-paving property* (EPP) if for every assignment of countable sets $c_x \subseteq X$ to points $x \in X$ there is $S \in \mathcal{U}$ with $y \notin c_x$ for all distinct $x, y \in S$.

Thus [3], Theorem 2.1 says normal measures have EPP (via the diagonal intersection), and Theorem 4.1 says $\mathcal{D} \otimes \mathcal{D}$ has EPP (via homogeneity). The point of the definition is that EPP is exactly what the positive theory of [3] consumes:

Theorem 5.2. *Let $\mathcal{U}$ be a κ-complete nonprincipal ultrafilter with EPP on an index set X, $|X| = \kappa$, and let $\Phi(\mathcal{U})$ be the block filter in $\mathcal{P}(\ell^2(X))$. Then every conclusion of [3], Sections 2–3 holds for $\mathcal{U}$: exact paving of every bounded operator on a measure-one block; the exact dichotomy for closed subspaces; the diagonal-flattening estimate; and $\Phi(\mathcal{U})$ is a proper, upward closed, $<\kappa$-intersection-closed, nonprincipal filter deciding every closed subspace — a κ-complete nonprincipal pluf — with $S_{\Phi(\mathcal{U})}$ a singleton and the round-slice property for every operator. In particular, all of this holds for $\mathcal{U} = \mathcal{D} \otimes \mathcal{D}$.*

Proof. Exact paving of a bounded operator T asks precisely for a mutually avoiding set for the assignment $c_x = \operatorname{supp}(Te_x) \cup \operatorname{supp}(T^*e_x)$, which is countable because vectors of $\ell^2(X)$

have countable support; this is EPP verbatim. Every subsequent argument of [3], Sections 2–3 — the orthogonality of the defect vectors $P_{W^\perp} e_x$ over the paving set, the exact computation $W \cap \mathcal{H}_{S_0} = \mathcal{H}_{S_1}$, the ultrafilter decision, the filter axioms (which use κ-completeness and, for nonprincipality, that countable sets are $\mathcal{U}$-small, immediate from κ-completeness and nonprincipality), the flattening (which uses paving, the existence of ultrafilter limits of bounded functions, and a termwise series comparison), and the singleton face (paving plus flattening plus the state estimates of [1]) — consumes normality only through the paving conclusion, as one verifies by inspection; indeed the mechanized development accompanying [3] already isolates the dependence, deriving the dichotomy, the flattening and the filter packaging from the paving statement together with κ-completeness-type hypotheses alone. Theorem 4.1 supplies paving for $\mathcal{D} \otimes \mathcal{D}$, and $\mathcal{D} \otimes \mathcal{D}$ is κ-complete and nonprincipal. □

Corollary 5.3 (Answer to [3], Question 5.5)**.** *Maximality of the block filter does not characterize normality up to isomorphism: $\Phi(\mathcal{D} \otimes \mathcal{D})$ is a κ-complete nonprincipal pluf with singleton face, while $\mathcal{D} \otimes \mathcal{D}$ is isomorphic to no normal measure (Proposition 2.3).*

The invariants now sit as follows, all implications for κ-complete nonprincipal ultrafilters on a set of size κ:

$$\text{full-selector} \iff \text{normal} \implies \text{EPP} \implies \Phi(\mathcal{U}) \text{ maximal} \implies \sigma\text{-Q},$$

where the last implication is [3], Proposition 5.3, and EPP implies σ-Q directly: given a countable-piece partition, apply EPP to $c_x = \sigma(x) \setminus \{x\}$; a mutually avoiding set is a partial selector. The product has EPP, maximality and σ-Q while failing the full-selector property; so maximality detects strictly less than normality. Whether the two middle implications reverse we do not know (Question 7.1).

Remark 5.4 (The two cardinals compared)**.** At $\aleph_0$, the block filter of an ultrafilter (completed by the finite-codimension subspaces) is maximal exactly for selective ultrafilters ([2], Theorem 3.4), and products $U \otimes V$ are never selective; at κ along κ-complete ultrafilters, products of a normal measure *are* maximal. The linear theory at measurable cardinals thus sees Rudin–Keisler structure more coarsely than the classical theory at ω: the class of "tame" ultrafilters strictly exceeds the RK-minimal ones. The mechanism is plain in the proofs: at ω, maximality needs the full Ramsey-theoretic strength of selectivity through Mathias's theorem, while at κ the countable-support phenomenon reduces maximality to paving, and paving to homogeneity — which, unlike normality, survives finite products.

6 Constraints on blockers at $\aleph_0$

We now return to the rigidity conjecture of [2]. Fix a non-selective ultrafilter $\mathcal{U}$ on $\mathbb{N}$, a witness partition $\mathbb{N} = \bigsqcup_k A_k$ (no piece and no selector in $\mathcal{U}$), the constraint vectors $f_k = \sum_{n \in A_k} 2^{-n} e_n$, and $W = \{f_k : k\}^\perp$; recall ([2], Theorem 3.4) that W is addable to the block filter $\Phi(\mathcal{U})$ but not a member. Call a closed subspace R an *addable blocker* if $R \cap W = 0$ and $\dim(R \cap \mathcal{H}_S) = \infty$ for every $S \in \mathcal{U}$; the rigidity conjecture ([2], Conjecture 3.8) asserts none exists, equivalently that W belongs to every pluf extension of $\Phi(\mathcal{U})$. The results of [2] exclude blockers spanned by disjointly supported families and blockers meeting a single piece-block two-dimensionally. Here are three further constraints. For $T \subseteq \mathbb{N}$ write $\mathcal{H}_T$ for the block, and for $x \in \mathcal{H}$ let the *piece-spread* of x be $\{k : x|_{A_k} \neq 0\}$.

Lemma 6.1 (Generalized thinness)**.** *If R is closed with $R \cap W = 0$, then for every finite set F of piece indices,*

$$\dim\Big(R \cap \mathcal{H}_{\bigcup_{k\in F} A_k}\Big) \;\le\; |F|.$$

Proof. The $|F|$ linear functionals $\langle\,\cdot\,, f_k\rangle$, $k \in F$, cut any subspace of $\mathcal{H}_{\cup F}$ of dimension exceeding $|F|$ in a nonzero vector v orthogonal to all f_k, $k \in F$; and $v \perp f_j$ for $j \notin F$ automatically (disjoint supports), so $v \in R \cap W = 0$. □

Taking F a singleton recovers [2], Lemma 3.6.

Lemma 6.2 (Support)**.** *If R is an addable blocker, then $T^* := \bigcup_{x\in R} \operatorname{supp} x \in \mathcal{U}$. Consequently no addable blocker is contained in $\mathcal{H}_T$ for any $T \notin \mathcal{U}$ — in particular, none is contained in the block of a partial selector, or of any union of a partial selector with a set outside $\mathcal{U}$.*

Proof. If $T^* \notin \mathcal{U}$ then $S_0 = \mathbb{N} \setminus T^* \in \mathcal{U}$; but $R \subseteq \mathcal{H}_{T^*}$, so a nonzero $x \in R \cap \mathcal{H}_S$ for $S \in \mathcal{U}$ has infinite support inside $T^* \cap S$. More precisely, for any $T \supseteq T^*$ with $T \notin \mathcal{U}$: for every $S \in \mathcal{U}$, $R \cap \mathcal{H}_S \subseteq \mathcal{H}_{T\cap S}$, and $\dim(R \cap \mathcal{H}_S) = \infty$ forces $T \cap S$ infinite; "$T \cap S$ infinite for all $S \in \mathcal{U}$" is equivalent, for an ultrafilter, to $T \in \mathcal{U}$ (otherwise $\mathbb{N} \setminus T \in \mathcal{U}$ and $T \cap (\mathbb{N} \setminus T) = \emptyset$). So $T^* \in \mathcal{U}$. □

Lemma 6.3 (Piece-spread)**.** *If R is an addable blocker, then for every $S \in \mathcal{U}$ the space $R\cap\mathcal{H}_S$ contains a vector of infinite piece-spread.*

Proof. Fix $S \in \mathcal{U}$ and write $R_S = R\cap\mathcal{H}_S$, a closed, hence completely metrizable, subspace with $\dim R_S = \infty$. The vectors of R_S of piece-spread contained in a fixed finite F form the closed set $E_F = R_S \cap \mathcal{H}_{\bigcup_{k\in F} A_k}$, of dimension at most $|F|$ by Lemma 6.1; a finite-dimensional proper closed subspace of an infinite-dimensional Banach space is nowhere dense. The vectors of finite piece-spread form the countable union $\bigcup_F E_F$ over finite F, which by the Baire category theorem is not all of R_S. □

Remark 6.4. Together with [2], Proposition 3.7, the lemmas confine any counterexample to the conjecture to a narrow shape: a blocker must have total support in $\mathcal{U}$ (Lemma 6.2), meet every finite union of piece-blocks in dimension bounded by the number of pieces (Lemma 6.1), be spanned by no disjointly supported family ([2]), and sustain, inside every measure-one block simultaneously, infinitely many dimensions of vectors each spread over infinitely many pieces (Lemma 6.3) — that is, its intersection with every $\mathcal{H}_S$ must be produced by cancellation among overlapping supports across infinitely many pieces, never by whole generators. The measurable-cardinal results of Sections 3–5 bound the conjecture from above: along $\mathcal{D}$ or $\mathcal{D} \otimes \mathcal{D}$ the σ-Q-point property holds, so no witness subspace exists and the rigidity question does not arise. Rigidity, if it holds, is a phenomenon of small cardinality, tied to the same Ramsey-theoretic poverty of ω that makes selectivity a special property there and a theorem at κ.

7 Questions

Question 7.1. For κ-complete nonprincipal ultrafilters on a set of size κ: does maximality of the block filter imply EPP? Does the σ-Q-point property imply EPP (equivalently, given

[3], Proposition 5.3 and Theorem 5.2, are σ-Q, EPP and maximality all equivalent)? A negative answer to either would require an ultrafilter separating selector-type from free-set-type combinatorics at κ; candidates presumably live among the many κ-complete ultrafilters of a Kunen-style model rather than among iterated products, which Theorem 4.1 and its remark place on the EPP side.

Question 7.2. Do mixed products $\mathcal{D}_1 \otimes \mathcal{D}_2$ of two distinct normal measures on the same κ have EPP? The homogeneity argument as given requires one set H serving both coordinates, i.e. Rowbottom-homogeneous sets in $\mathcal{D}_1 \cap \mathcal{D}_2$, which need not exist. A proof or a counterexample would locate how much of Theorem 4.1 is about products and how much about self-products.

Question 7.3. The rigidity conjecture of [2] stands: does some non-selective $\mathcal{U}$ admit an addable blocker of its witness subspace — necessarily of the shape delimited in Remark 6.4 — or does W belong to every maximalization of every block filter of a non-selective ultrafilter? A first target consistent with all known constraints: a blocker for a product ultrafilter $\mathcal{U} = V \otimes V$, built from vectors spread along the graphs of finitely-branching trees of pieces, where the selector obstructions of [2], Proposition 3.7 do not directly apply.

Formal verification

Every theorem of this paper has been formalized and machine-checked in Lean 4 with Mathlib (toolchain v4.28.0), extending the artifacts of [3]: Lemma 2.2, without countability or nonemptiness of the pieces; Proposition 2.3, through an isomorphism-invariance lemma for the full-selector property; Theorem 3.1 and Corollary 3.2; Theorem 4.1, as the enumeration of twelve ordered placement patterns, of which four are the mid-region patterns treated by the tail-then-pivot step of Section 4; the exact-paving property with the implication EPP $\Rightarrow$ σ-Q; the lattice-level clauses of Theorem 5.2 — operator paving, the exact dichotomy, the flattening estimate, and the decision property and nonprincipality of the block filter, instantiated at the Fubini product — and Lemmas 6.1–6.3. Classical inputs are quarantined as hypotheses (Rowbottom homogeneity for triples and quadruples, membership of tails, uncountable pivots, and the Fodor property), so every mechanized statement is a theorem of ZFC. The injectivity theorem consumes the tails hypothesis, which holds for every uniform ultrafilter and is invisible at a normal measure. The artifacts compile with no unproved obligations, and every statement's axiom footprint is exactly `propext`, `Classical.choice`, `Quot.sound`. Two items are not covered: the finite-power remark of Section 4, and the state-theoretic clauses of Theorem 5.2 (singleton face, round slices). The face machinery those clauses invoke is itself verified in [1]; its transfer to ultrafilters given only by EPP is not part of the present development.

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References

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